

WEEK 3 – LECTURE 5

Artificial Intelligence Foundations

Uninformed Search Algorithms

Breadth-First Search (BFS) & Depth-First Search (DFS)

CSE276 | Unit II



Quick Recap



Today's Objective:

How do we actually *search* this space to find the solution?

Motivation: Where to Go First?



Google Maps

Exploring possible roads



Robot

Searching for an object



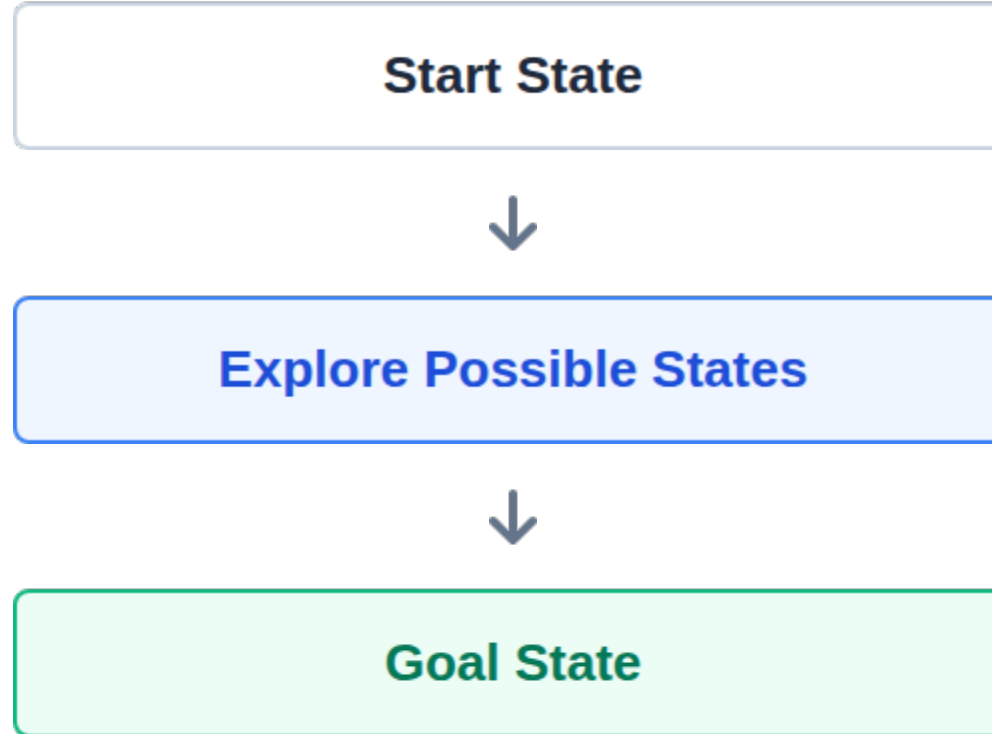
Maze Solving

Finding the exit path

How should AI decide which path to explore first?

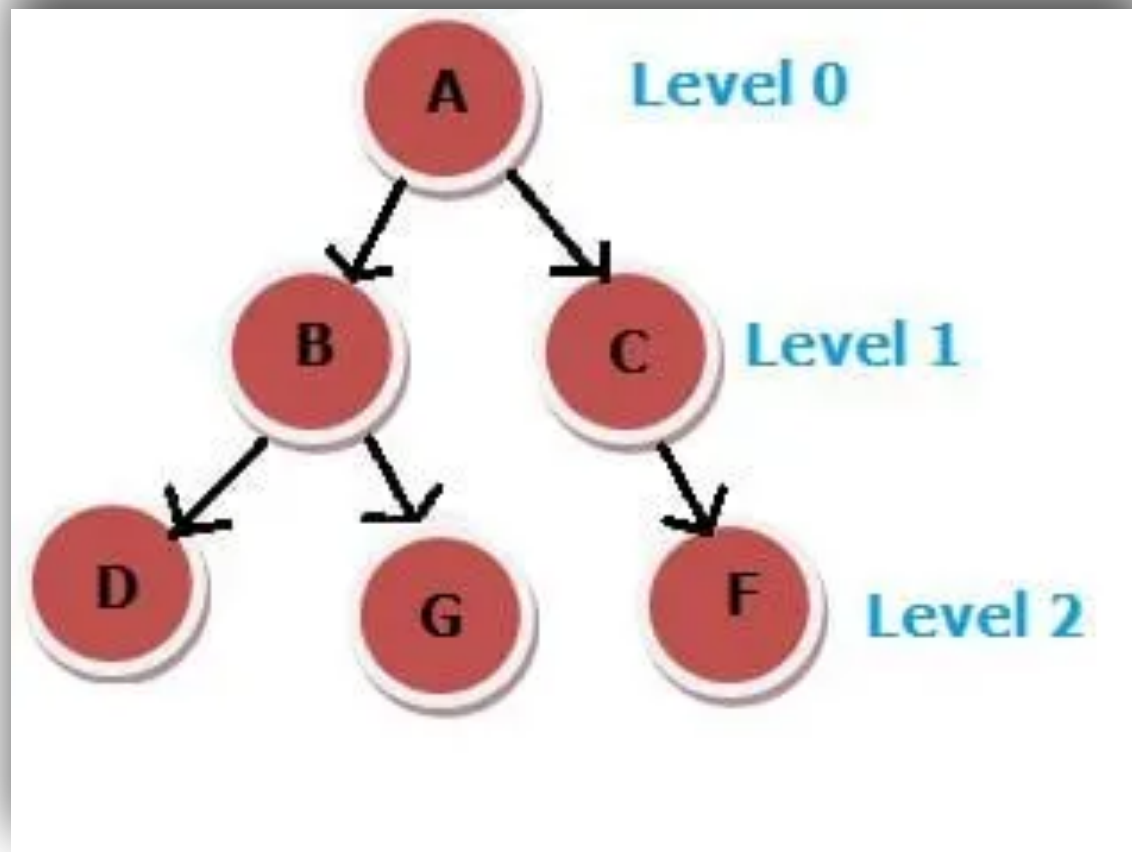
What is Search in AI?

The process of systematically exploring possible states to reach a goal.



Search Tree Representation

- **Node:** Represents a state.
- **Edge:** Represents an action.
- 👤 **Root:** The initial state.
- 👤 **Children:** Subsequent possible states.
- 🚩 **Goal Node:** The solution state.



What are Uninformed Search Algorithms?

Also known as "Blind Search".

The AI has no additional information about how far the goal is. It only knows how to generate successors and check if a state is the goal.

AI without hints



Explore systematically



Find Goal

Everyday Analogy: Searching a Mall

You are looking for a specific store in a multi-level shopping mall.



Method A

Visit every shop on the 1st floor.
Then visit every shop on the 2nd floor.
Then the 3rd floor...



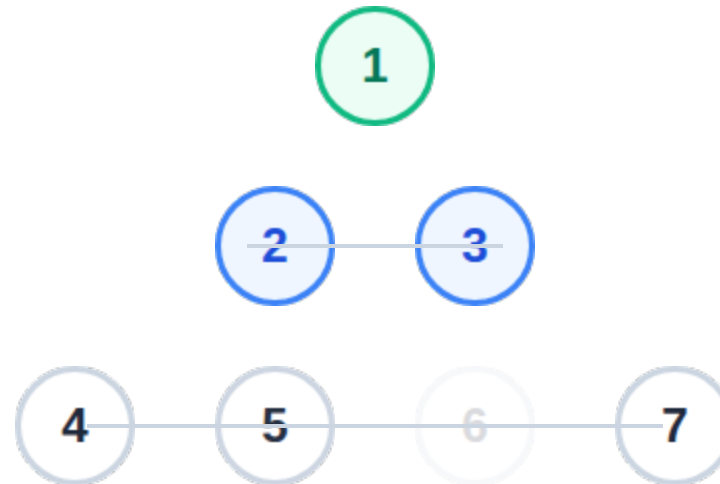
Method B

Keep walking down one long corridor until the very end. If it's not there, turn around and try the next corridor.

Which method represents BFS? Which is DFS?

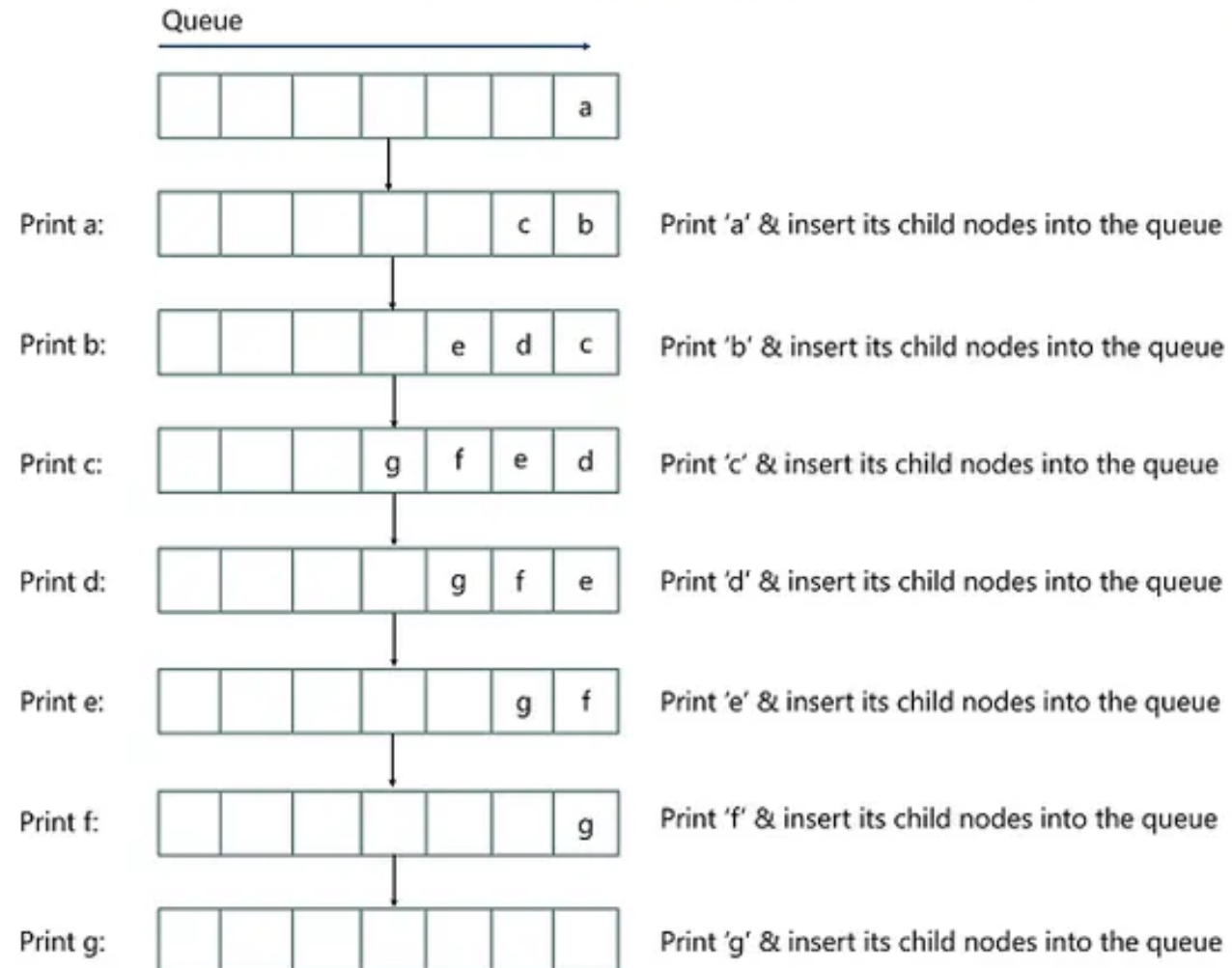
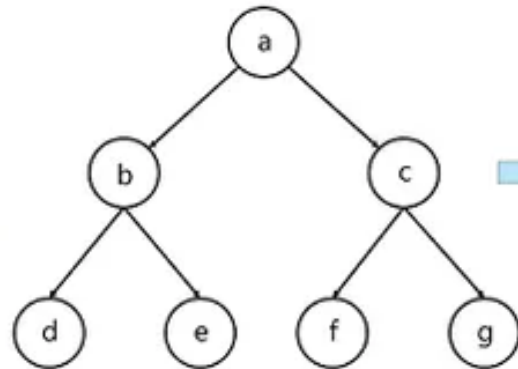
Breadth-First Search (BFS)

Explores the search space **level by level**.



Expands all nodes at depth d before depth $d+1$.

BFS Example



BFS Characteristics



Strategy

Level by level exploration.



Completeness

Yes. Will find a goal if one exists.



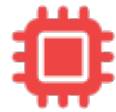
Shortest Path

Yes (for unweighted graphs).



Data Structure

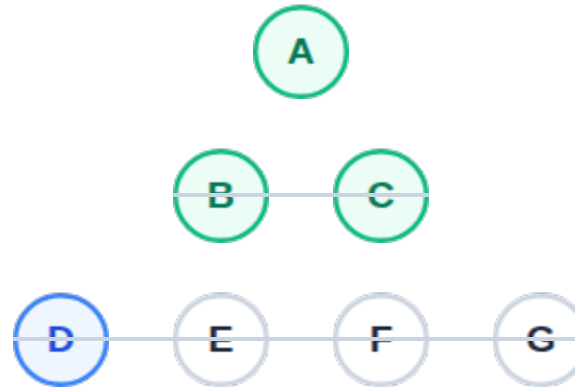
Uses a **Queue** (FIFO).



Memory

High. Must store all nodes in current level.

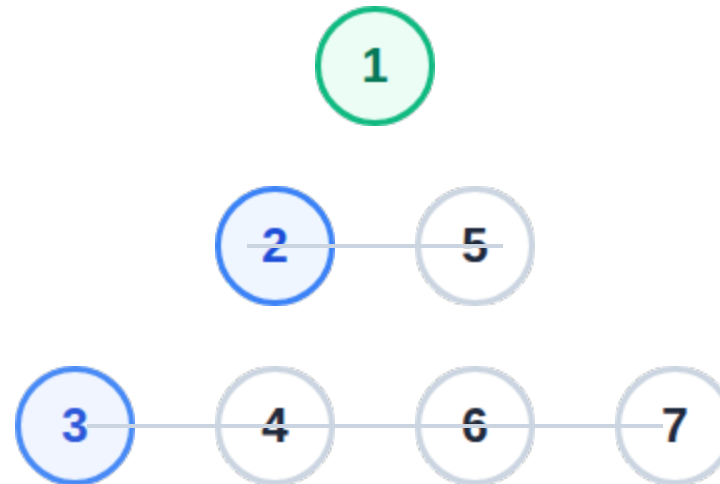
Interactive Activity



If we are using BFS, which node will be visited next?

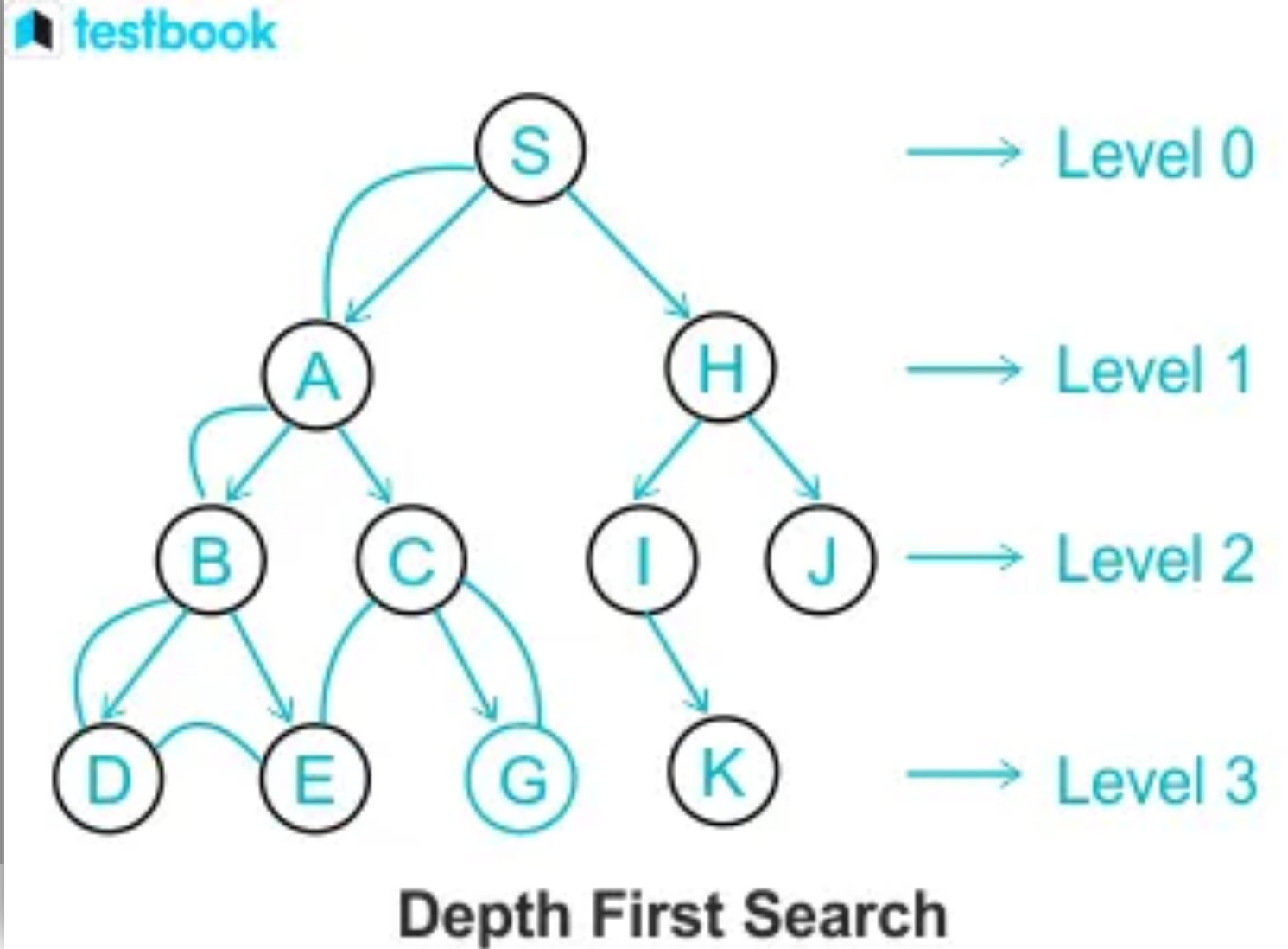
Depth-First Search (DFS)

Explores as **deep as possible** before backtracking.



Follows one path to the end, then backs up.

DFS Example



DFS Characteristics



Strategy

Deep dive, then backtrack.



Completeness

Yes (in finite spaces),
No (in infinite).



Shortest Path

No guarantee.



Data Structure

Uses a **Stack** (LIFO)
or Recursion.



Memory

Low. Only stores
current path.

BFS vs DFS

Feature	E↔S	E↓S
Exploration Style	Level by level (Broad)	Deep dive & Backtrack
Data Structure	Queue (FIFO)	Stack (LIFO) / Recursion
Memory Usage	High (Stores levels)	Low (Stores current path)
Shortest Path?	Yes (Unweighted)	No

Real-World Applications

↔ Use BFS for:

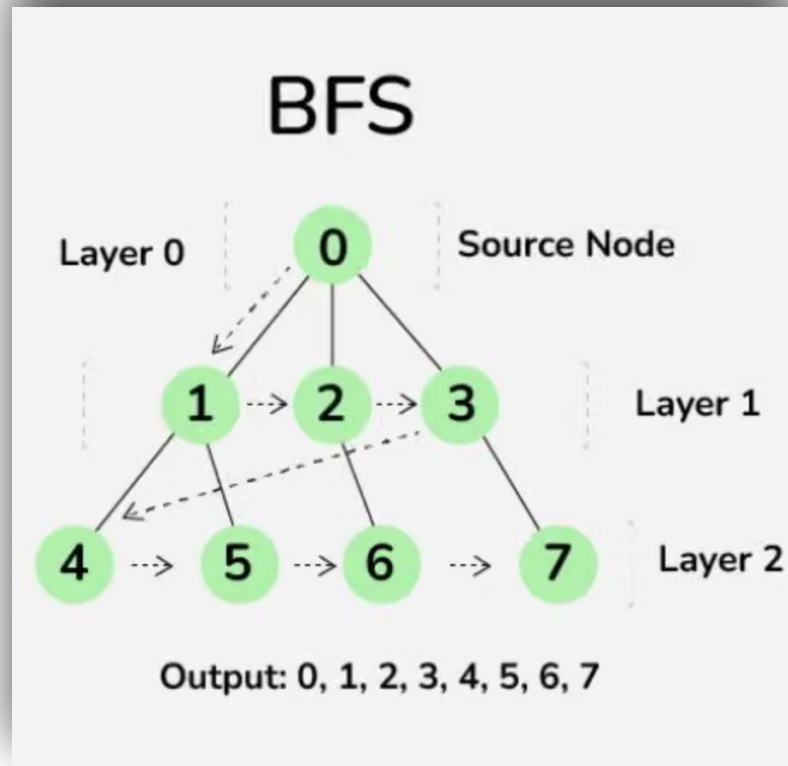
-  Finding shortest route in maps.
-  Social network friend suggestions (1st degree, 2nd degree).
-  Network broadcasting.

↓ Use DFS for:

-  File system directory traversal.
-  Solving mazes & puzzles.
-  Website crawling.

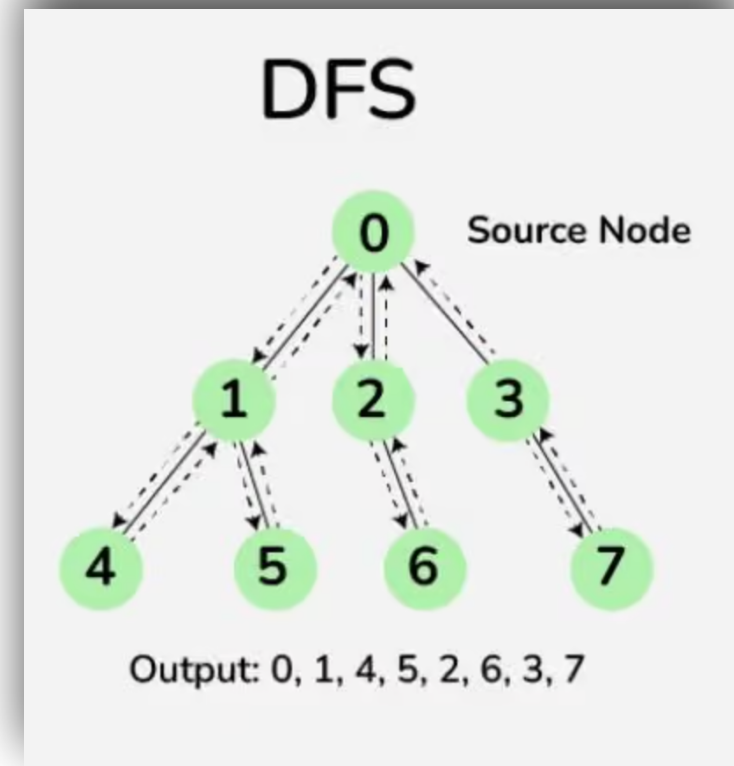
Case Study: Maze Solving

BFS Exploration



Expands evenly like water.

DFS Exploration



Darts to dead ends, then backtracks.



Board Activity

Manual Traversal of a Graph

We will trace the exact order nodes are visited for both BFS and DFS.

Common Misconceptions



"BFS is always faster than DFS."

Not true. If the goal is deep, DFS might find it instantly.



"DFS always finds the shortest path."

DFS finds a path, but rarely the shortest one.



"DFS never backtracks."

Backtracking is fundamental to how DFS works when it hits a dead end.

Key Takeaways



BFS explores level by level
(Queue).



DFS explores depth first
(Stack).



BFS guarantees shortest
path.



DFS uses less memory.

Quick Quiz

Which data structure is required to implement Breadth-First Search (BFS)?

- ☐ A) Stack
- ☐ B) Queue
- ☐ C) Array
- ☐ D) Hash Map

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Preview: Next Lecture & Lab

Upcoming Lab:

-  Implement BFS & DFS in code.
-  Compare their execution times.

Next Lecture: Informed Search

-  What if AI has hints?
-  A* Search Algorithm.

